

OPERATIONS & CUSTOMER EXPERIENCE ANALYSIS

Stabilize the network before scaling the footprint.

An evidence-based operating diagnosis of peak-hour failure concentration, Gen1 charger underperformance, and network-wide rider churn.

Executive decision signal: Redirect the next operating budget toward targeted battery buffering and Gen1 modernization; pause geographic expansion until reliability and churn recover.

THE DECISION IN ONE PAGE

Operational friction—not pricing or footprint—is the constraint.

VoltRelay’s growth is masking an operating model that breaks at the exact hours when gig-economy riders depend on the network most. The evidence points to concentrated hardware and inventory bottlenecks, not a broad-based demand or market-fit issue.

94.04%

TOP-LINE SUCCESS RATE

>200k

CRITICAL FAILURES

11.71 min

AVERAGE WAIT BEFORE ABANDONMENT

~40%

TRUE RIDER CHURN

“The network is scaling volume while degrading the rider experience that makes the volume repeatable.”

What is happening

Failure events cluster in the 12 PM lunch rush and 7–9 PM evening window. Gen1 cabinets fail at 6.51%, materially worse than Gen2 and Gen3 at roughly 4.1%.

What leadership should do

Prioritize physical throughput: add targeted battery inventory at Gen1 hubs, evaluate retrofits, and defer Wave 3 expansion until churn is below 20%.

Problem Understanding

Scale has increased swaps and revenue, but the network's operating health is deteriorating. The strategic question is where the next budget dollar creates the fastest reliability and retention recovery.

Decision context: Leadership must choose among geographic expansion, deeper battery inventory, network-wide pricing changes, or fleet-partner exclusivity—without mistaking top-line growth for a healthy operating model.

Observed symptoms

- Rising service failures during predictable demand spikes.
- Eroding per-swap margins as failed attempts and rider abandonment increase.
- Declining new-rider retention despite continued network growth.
- Risk that expansion will replicate the same bottleneck in new zip codes.

Management questions the analysis must answer

Where is the break?

Which hours and stations create the highest operational and customer-experience damage?

What is causal?

Is churn primarily a pricing, geography, equipment, inventory, or rider-segment issue?

What is scalable?

Which intervention improves throughput before capital is committed to footprint growth?

What should be measured?

Which leading indicators should govern budget release and expansion readiness?

Analytical Approach

The investigation combined 18 months of operational history with rider-level retention logic and equipment-level reliability comparisons.

18 mo

OPERATIONAL HISTORY

3.9M

SWAP EVENTS

20k

RIDER PROFILES

6

CITY NETWORK

1**Data integrity resolution**

Corrected a timestamp logging error linked to firmware v3.2.0, deduplicated offline-batch sync anomalies, and standardized geographic market naming conventions.

2**KPI definition**

Redefined rider churn at the individual user level—30 days of inactivity—rather than the transaction level to eliminate survivorship bias.

3**Hypothesis testing**

Compared outcomes across vehicle class (2W vs. 3W), commercial arrangement (Fleet vs. Independent), geographic market, and equipment generation.

Analytical implication: A network average is not sufficient to diagnose the operating model. The analysis intentionally looked for concentration by time of day and charger generation while testing whether churn varied meaningfully across rider segments.

Interpretation standard

Use the smallest operational unit that can explain the customer outcome: hour × station × charger generation, linked back to rider-level inactivity rather than transaction counts alone.

Key Insights

Four findings explain why the network can look healthy at the top line while still failing the core rider promise.

1. The mealtime bottleneck is the operating failure.

The 94.04% success rate conceals more than 200,000 critical failures. Failure volume concentrates in the 12 PM lunch rush and the 7–9 PM evening commute—precisely the windows when riders need predictable uptime.

2. The wait-time cliff destroys rider economics.

When the network bottlenecks, riders wait an average of 11.71 minutes before abandoning the attempt. For logistics and quick-commerce riders, this is lost earning capacity, not merely inconvenience.

3. Gen1 is the reliability deficit.

Gen1 cabinets fail at 6.51% versus 4.11% for Gen2 and 4.10% for Gen3. Gen1 accounts for 118,531 absolute failures—more than all newer equipment combined.

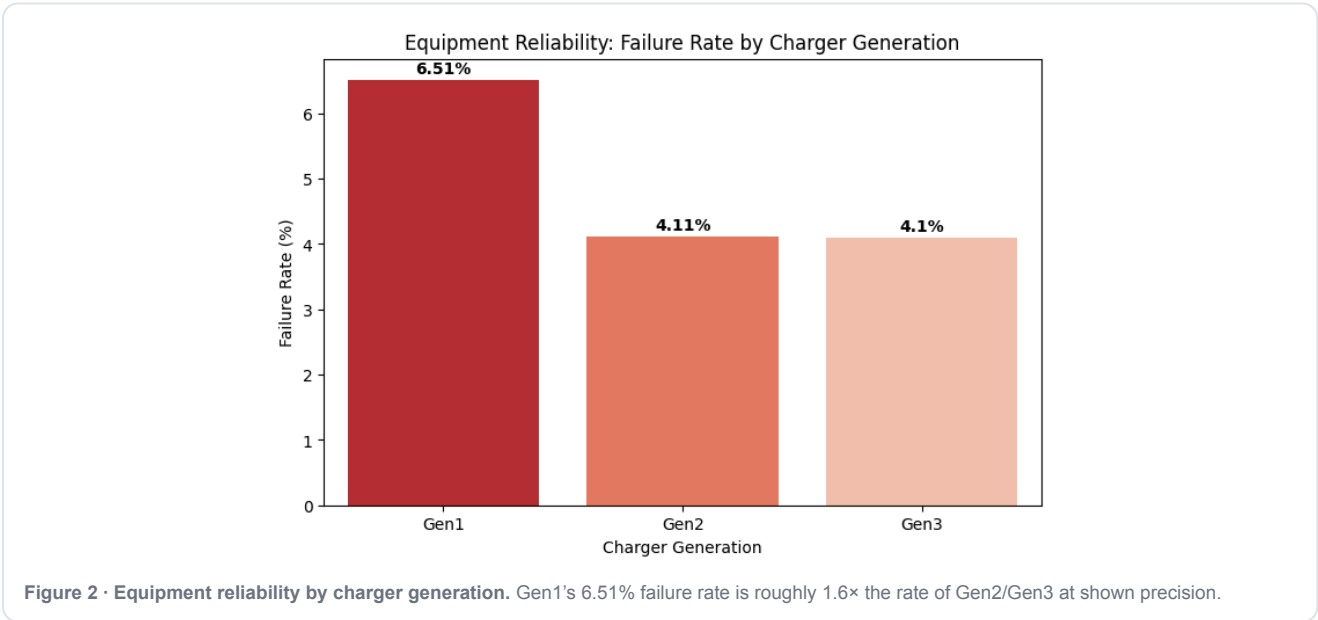
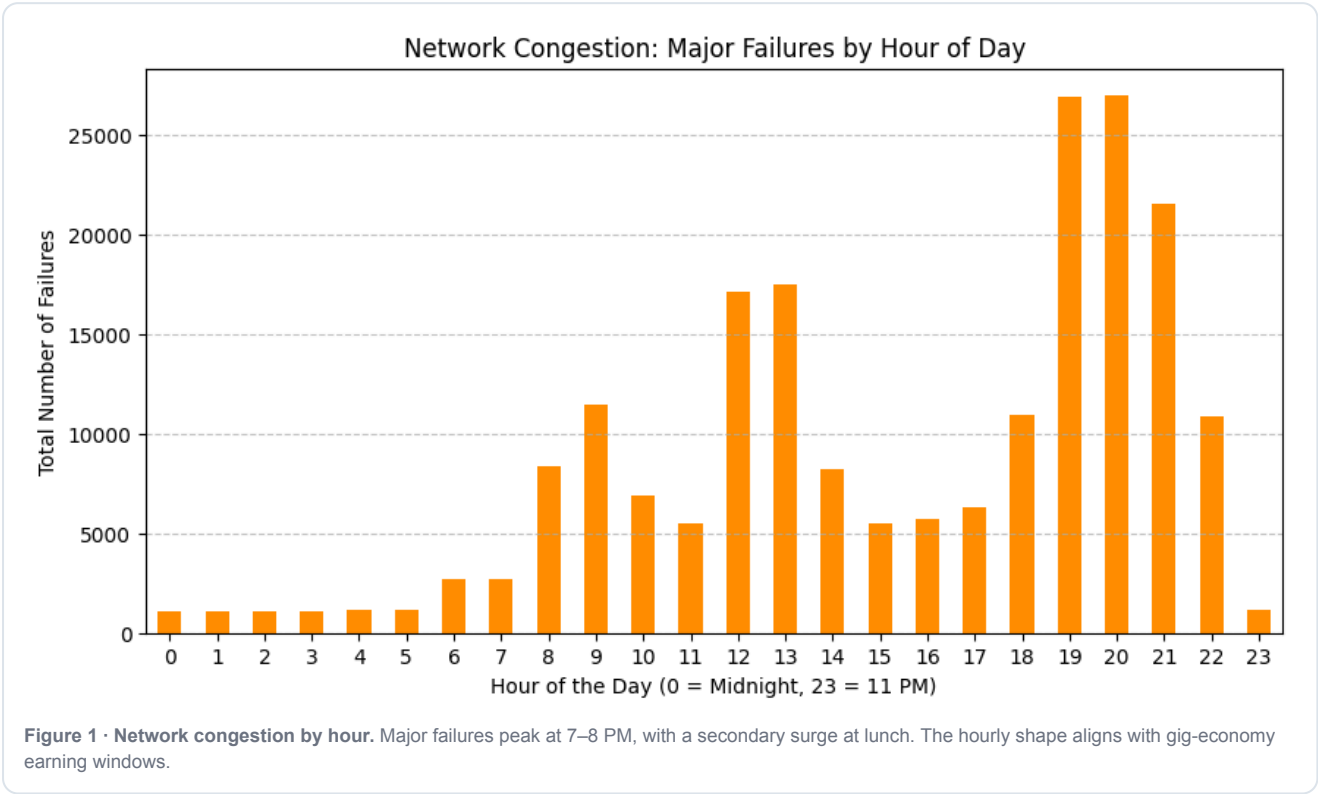
4. Churn is universal, not localized.

True churn is approximately 40% across cities (37–42%), vehicle classes (39.4% for 2W; 41.5% for 3W), and rider segments (40.0% Fleet; 39.2% Independent).

Headline: The evidence is consistent with a physical throughput problem that manifests as customer attrition across the entire network—not a narrow issue isolated to one city, rider type, or pricing tier.

Supporting Visualizations

The figures below make the concentration and equipment gap visible: failures are time-bound, and Gen1 is the clear reliability outlier.



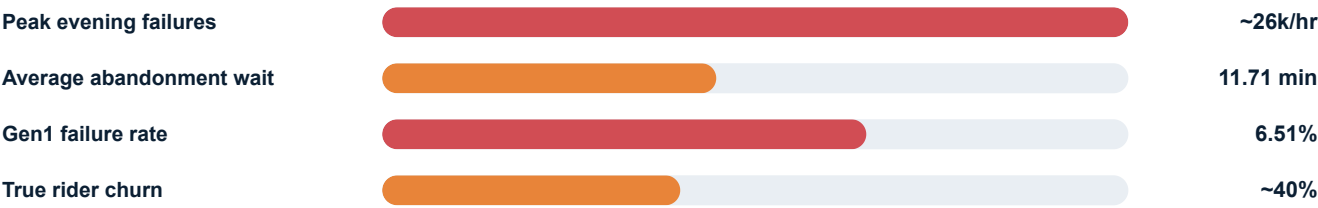
Equipment	Swap attempts	Major failures	Failure rate	Signal
Gen1	1,820,752	118,531	6.51%	Primary reliability deficit
Gen2	1,681,849	69,124	4.11%	Improved baseline
Gen3	372,398	15,267	4.10%	Comparable to Gen2

Business Findings

The business is not facing a broad demand problem. It is facing a reliability constraint that becomes visible during the highest-value operating windows.

Primary finding: Rider attrition is a predictable outcome of severe hardware bottlenecks during peak hours. Gen1 cabinets cannot process depleted batteries fast enough to support the throughput required by the core user base.

Evidence chain



What the data rules out

- Pricing as the primary churn lever.
- One-city or one-segment concentration.
- Battery degradation as the dominant explanation.

What the data supports

- Peak-hour throughput failure.
- Gen1 cabinet underperformance.
- Physical inventory and modernization response.

Strategic implication: Expanding the footprint before stabilizing Gen1 turnaround times will scale the failure pattern into new zip codes and increase the cost of recovery.

Actionable Recommendations

Sequence investment around throughput recovery. Use a targeted operating response first, then release capital against measured reliability improvement.

1. Primary budget allocation — add inventory where demand is hardest.

Immediately deploy heavy battery inventory into high-density transit hubs using Gen1 equipment. Build a deep physical buffer that absorbs lunch and dinner spikes while older chargers catch up.

2. CapEx reallocation — evaluate Gen1 retrofits.

Run a 12-month cost-benefit comparison between localized battery buffering and retrofitting Gen1 cabinets with Gen3 internals. Shift capital toward modernization where the retrofit has the lower cost per recovered failure avoided.

3. Halt geographic expansion — pause Wave 3.

Hold the Wave 3 station expansion plan. Concentrate operating resources on restoring network reliability and reducing true churn from approximately 40% to below 20% before scaling the footprint.

Recommended management scorecard

Measure	Current signal	Decision threshold	Owner cadence
Gen1 failure rate	6.51%	Trend below Gen2/Gen3 gap	Weekly
Peak-hour major failures	~26k/hr evening	Sustained step-down at top hubs	Daily
Abandonment wait	11.71 minutes	Material reduction during peak windows	Daily
True rider churn	~40%	<20% before expansion	Monthly

Do not lead with price or footprint.

Pricing pilots and new stations cannot compensate for stockouts and slow charging turnaround. Fix the physical bottleneck first; treat expansion as a reward for reliability recovery, not a substitute for it.

Decision Summary

The operating decision is straightforward: protect the earning window, modernize the worst equipment, and earn the right to expand.

Recommended sequence		
Phase	Focus	Expected business effect
0–90 days	Battery buffering at Gen1 peak hubs	Reduce stockouts, abandonment, and lost rider earnings during demand spikes
90–180 days	Retrofit / modernize Gen1 stations where economics justify it	Lower structural failure rate and improve peak-hour throughput
Until proven	Pause Wave 3 expansion	Avoid replicating a broken operating model before churn recovers

Invest in reliability where riders feel the failure most. Expansion should follow proof—not precede it.

Source: client-provided narrative, operating metrics, and supporting figures. This report reformats and synthesizes the supplied analysis; it does not introduce new operational data.